Tracking Changes in Buildings over Time

Fully Automated Reconstruction and Difference Detection of 3d Scan and BIM files

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Architectural and Engineering Communities are interested in the detection of differences between different representations of the same building. These can be the differences between the design and the as-built-state of a building, or the detection of changes that occur over time and that are documented by consecutive 3D scans. Current approaches for the detection of differences between 3D scans and 3D building models are however laborious and work only on the level of a building element. We demonstrate a novel highly automated workflow to detect differences between representations of the same building. We discuss the underlying tools and methods and the ways to communicate deviations and differences in an appropriate manner and evaluate our approach with a rich set of real world datasets.

Keywords: *3d scan, BIM, Machine learning, Point Clouds, Big Data*

INTRODUCTION: WHY ARE CHANGES IN BUILDINGS OBSERVED?

The monitoring of the progress on building sites, the control of geometric deviations between planned and as-built geometry and the check of planarity of building elements during construction planning are essential tasks in AEC. Further applications are located in the area of Facility Management and planning of renovation tasks, where the detection of differences between a documented state and the current state are a prerequisite for decision making and design, and where the quality of a 3D model that represents the current state of the building has to be assessed. The currently fastest and most precise way to document the geometry of an existing building is by means of Point Clouds, which are generated by 3d laser scanners or photogrammetric devices.Point clouds consist of millions of unrelated coordinates. This is diametrically opposed to BIM, the current state of the art paradigm for architectural planning and facility management, which represents relational object oriented data in a hierarchical structure.

Hence, finding differences between these two types of datasets first of all necessitates to represent

them in the same digital model and the same 3D coordinate system, i.e. to overlay point cloud data and BIM objects or to create a way to semantically link them.

Secondly it has to be understood, what types of differences shall be detected among a set of representations of the same building. Differences of interest can theoretically be observed on any level of granularity in a building, from the extension of a building to small changes in surface ornament. A huge search space is emerging, due to the complexity and size, that is inherent to buildings and therefore as well its representation. In order to filter and express relations between the many different elements the building profession has hence developed the concept of scales. This concept is fundamental to the structuring of Building Information Models, where a high level description of a building shape is in depended scales broken down into the smallest constituting parts and relations between the scales can be expressed in multiple ways.

This scalar and relational understanding of parts and whole provides a powerful approach to start and structure the search for differences. It allows for a discrimination between high level building related elements such as walls, floors, ceilings, depended elements, such as doors, windows and even smaller parts and elements, which are not related, such as furniture or other temporary elements. And it allows to search for differences on one level of detail and express found differences on a higher one, granting overview of e.g. changes in the whole building, and impact on related elements on a smaller scale in BIM. This goes along the interest of professionals, who will be interested in questions: which walls or windows have been removed, were added or changed in dimension. Here it is important to acknowledge, that the 3d scans are only capturing a certain moment in time and that most building elements are static in size and position. The "move" of wall by one meter, or the change in the dimension of a window is in reality a removal of one and an addition of another element. An algorithm might hence observe, that a window is at the same position but in a different size, or that a similar sized wall is in close vicinity to a previous existing one. The essence of a difference detection is concerned about missing or added elements.

Stakeholders will as well be interested in an overview, where in a building these changes take place and are eventually amassed, and where the most severe ones take place (Figure 1). Other differences of interest, as on the level of building services (Ducts, Pipes etc.), can as well be expressed on the level of objects. And while a tracking of differences of these is possible when they are exposed, image based methods do not work, when they are hidden. However, a professional might deduct that hidden elements changed, when the geometries of related elements, such as the wall they are enclosed in, change.

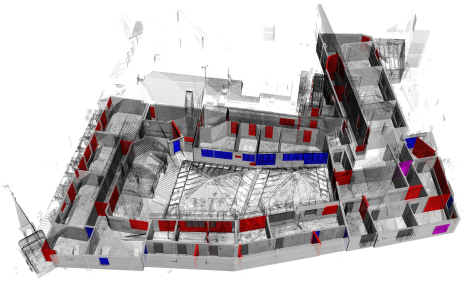


Figure 1
An an analysis between a 3d scan and a modelled BIM reveals missing elements (red) and elements with difference in size (blue) and type (window instead of door or vice versus - pink). Source: LE34.

We argue for a difference detection between building data, which takes point of departure in the most common method of how this type of data is structured - as building elements. This allows for more advanced types of query across scales and provides an effective filter for clutter (furniture, scanned persons etc.), as all of these can not be associated to building elements. For the difference detection we are looking first of all, whether elements have been removed or added between two or more representations of a building. Related questions, whether an element has been eventually moved or resized, whether it has only be partially removed or how much a element deviates from a planned or previous state are steps for subsequent detailed analysis.

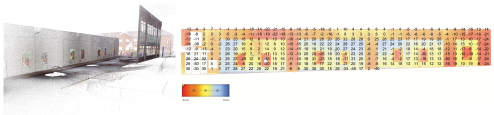


Figure 2
The detection of deviations to a plane is a typical use case, that takes place on the level of a single building element. Here the deviation of an outdoor wall (left) to the vertical plane. The result is visualised in a color gradient (right) and the deviations are indicated as numbers on an overlaid coarse grid.

STATE OF THE ART

Current approaches for a difference detection between Point Cloud data from photogrammetry (Tuttas 2014) or 3d Laserscan (Bosché 2010) and 3d geometry in form of 3D CAD or BIM data concentrate usually on the level of a single building element.

Related commercial implementations are available in software such as CloudCompare (Cloud-to-cloud distance), Bentley Pointools (Clash Detection and Differencing), PointSense (Clash Detection, Deformation Analysis), Geomatic (Automated Accuracy Analyzer), ClearEdge3D EdgeWise (Quality Assurance Tool). Research indicates an extension of this approach in order to identify e.g. the correct positioning of a door (Anil 2011). Common to the workflows in all these implementations is, that users are tasked to more or less manually correlate a Point Cloud and BIM model of a building in a common 3D space, identify a single building element and start an algorithm, which searches e.g. for the distance of points in the Point Cloud to the elements. Upon completion users receive information of differences and deviations (Figure 2) for this particular element. This workflows is overall very laborious.

Less attention was given to point-cloud- to -point-cloud difference -detection. This area is currently mainly investigated for the field of terraforming (Girardeau-Montaut 2005) or as general computer science related problem in the case of real-time 3d scanning, where an example of an implementation is available in the PCL library. However these approaches do per se not answer the questions on building element level, which stakeholders from the building profession have. None of the commercial implementations is able to provide an overview of all the differences, that have been detected and to express the quality of these in numerical terms, as for instance in a measure of overall accuracy. Research prototypes (Tuttas 2014) use octree space partitioning on correlated Point Cloud and BIM models in order to create a search space, where differences can be found.

Another principal approach (Tamke 2015), is to take a point of departure in the BIM model and search iteratively in the correlated point cloud for points in vicinity to each surface of e.g. a wall (Figure 3). The visual representation of all remaining points provide a good overview and indicate areas that require further inspection. Elements, which are not present in the BIM file can be detected (in the figure e.g. not modelled suspended ceilings in corridors).

Both approaches are able to provide a general overview of differences and indicate elements, which are modelled in BIM, but miss or deviate substantially in the point cloud. They are however not able to discriminate sufficiently clutter and detect added elements, such as walls, nor do they take into account, that due to the use of vision based technologies building elements can be overshadowed by other elements.

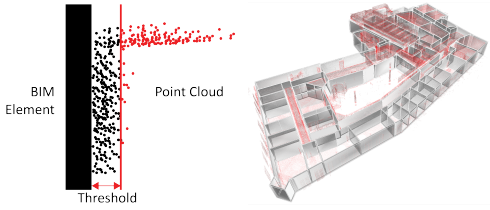


Figure 3
Difference detection through iteratively search of points in a point cloud, that are outside a certain distance from each BIM elements (red).

WORKFLOW FOR DIFFERENCE DETECTION BETWEEN POINT CLOUDS AND BIM

We propose a workflow to detect differences and deviations between interior representations of buildings in BIM and point cloud formats which operates on the level of building elements. It's point of origin lies in technologies developed in the EU funded DURAARK project (Beetz et al. 2016), where tools were developed, which are able to reconstruct a BIM model from point clouds (Ochmann 2016). These

tools reconstruct ca. 90% of the interior building elements found in BIM on LOD2 (Tamke et al. 2016) faithfully. We consider this amount sufficient for our research prototype. Being able to reconstruct BIM from point clouds provides an efficient means to compare Point Clouds of the same building at different times through the comparison on the base of a comparison of the two derived BIM models.

Our approach towards the Difference Detection of Point Clouds and BIM consists of four distinct steps:

1. Registration: Orienting the representations (point clouds and BIM models)
2. Association: Subsets of points are associated to either subsets of points or building elements depending on whether the tool is used for point-cloud-to-point-cloud or point-cloud-to-BIM association. During this step, unnecessary information such as furniture or scanned people, is excluded from the analysis. The focus is set on the building itself and it has to be checked particularly for false negative detections.
3. Data Analysis: The difference between the datasets is calculated
4. Refinement and user feedback: The results of step 3 are verified and refined and qualitative and quantitative feedback is provided to the users.

In our approach we try to achieve a great extend of automation and hence efficiency in each step, eventually trading in precision for speed.

Step 1: Automated registration of point-cloud and BIM models

The registration process is roughly based on work by Makadia et al. (Makadia et al. 2006) who achieve a coarse pre-alignment using correlations between Extended Gaussian Images in the Fourier domain, and subsequently perform a finer registration using the Iterative Closest Point algorithm (ICP).

Taking the assumption that the "up" direction is the z-axis and that most wall and/or structures are respectively vertical or horizontal, we can restrict ourselves to a successive alignment in the two-dimensional domain.

Vertical planes in both representations are extracted and their dominant normal directions are determined by clustering the directions of projections of the extracted planes into the xy-plane. The largest cluster of directions is then considered to be the most dominant direction, and the second dominant direction is constructed orthogonally to this direction. In case of BIM models, the planes are extracted directly from the surfaces of wall entities; in case of point clouds, a Random Sample Consensus (RANSAC) plane detection is performed, see (Schnabel et al. 2007) for further details. The dominant directions in both datasets provide hints for possible rotational alignments of the datasets around the z-axis, i.e. we assume that aligning the dominant directions to each other yields a valid rotational alignment of the datasets, which sufficiently restricts the search space in the rotational domain.

For each possible rotation around the z-axis, the projections of vertical planes along the z-axis are rendered as lines; one per building representation. Then the alignment maximizing the overlap between the two resulting line images is found using a Fourier transform based approach. The computed image alignment transformation is then transferred back into world coordinates and applied to the building representations. This results in an alignment of the datasets in the xy-plane. For determining the translational alignment along the z axis, a similar approach is used: Horizontal planes that potentially correspond to floor surfaces and vertical planes are rendered into an image, viewing the building representations from the side. Aligning the images to each other then results in a translational offset which is applied to the building representations (Figure 4).

The output of this first step is an RDF file (RDF transformation file), which describes the transformation that is necessary to achieve the registration of the two models.

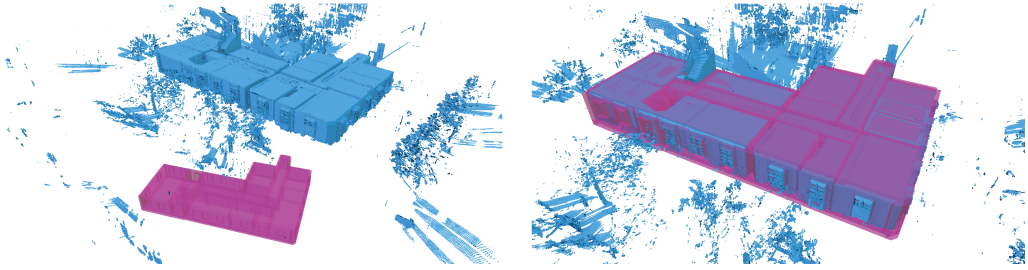


Figure 4
Step 1: Automatic alignment. Different representations (here BIM shown in cyan and point-cloud shown in blue) are automatically aligned in the three dimensional space.

Step 2: Association of point subsets to IFC building elements

Since our proposal for a difference analysis is based on building elements, we use a technique developed in the DURAARK project (IFC reconstruction) in order to detect building elements, such as rooms, floors, ceilings, walls, doors, and windows along with their relations in a Point Cloud (Figure 5). In the following we will briefly outline this method, for further details we refer to (Ochmann 2016).

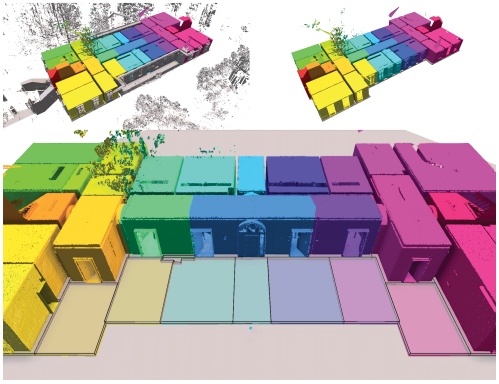


Figure 5
The second step associates Points to building elements (room, wall, floor, ceiling).

In a first step a coarse segmentation of the point cloud separates rooms and outlier points, filters out clutter outside the building. Infinitely long vertical wall candidate planes are generated by means of a RANSAC algorithm (Schnabel et al. 2007), which yields candidates for wall entities that are shared between rooms. The determination of a globally plausible configuration of connected wall elements is then posed as a graph labeling problem, which is solved using an energy minimization approach. Openings are detected and classified as doors and windows by means of supervised learning.

An intermediate data structure of the process is the representation of a building story as a halfedge graph in which faces are interiors of different rooms and edges are walls between adjacent rooms. The neighborhood relation of rooms is also directly contained in this structure, i.e. two rooms are neighboring if and only if they share a common edge. This representation allows for fast queries to which high-level building element (room, wall, floor, ceiling) a given 3D point belongs. This allows us to associate all measured points of a point cloud to the aforementioned building elements.

The resulting BIM model is stored in IFC format. Furthermore a RDF file is created. This RDF file holds the information on relationships between a point cloud and a BIM file. The association is expressed as pairing of point cloud subsets with respective BIM geometry, and the UUID number of the particular geometry indicates the association. Points in the point cloud, which are not associated to building elements, can be determined by a simple boolean operation.

Step 3: Data Analysis

The aim of the data analysis is to detect and visualize the differences between two or more building datasets. Our approach uses first of all the UUIDs of building elements, which were either provided by the users as BIM file or were generated in the previous steps from point clouds. Specifically we are using

the generated RDF files. The associated point cloud data is in a subsequent step used for verification and refinement of the initial analysis.

To perform the Difference Detection, we have developed a custom Grasshopper components library, which is based on the Volvox plugin for Rhino/Grasshopper (Zwierzycki et. al 2016). Thanks to Grasshopper's visual programing interface, the feedback is instantaneous and detection of changes can be visualized. All above takes place in a custom Grasshopper components library utilizing partially linked C++ APIs.

Before the actual analysis, data from the RDF files has to be interpreted. This takes place in the following process:

1. Parsing of the RDF transformation file - this file contains transformation matrices assigned to UUID identifiers which reference either particular points in the point cloud or building elements in the BIM model.
2. Parsing of the RDF association file - the association file contains a list of indices of points from the point cloud, tagged with an UUID identifier which connects them with particular part of BIM geometry.
3. Transformation and splitting of the point clouds - based on the former files, the point clouds collection can be oriented in space to match the BIM model, after which it is split into parts based on the RDF association file.

The interpretation of data is based on UUID identifiers of each element. The collection of the UUID identifiers is analyzed concerning building elements unique to each data set, building elements common for all datasets, building elements appearing in several datasets etc. The analysis is performed on sets of UUID identifiers using simple set theory logic. Not matching UUIDs for the same element in different BIM files are identified through a control of geometric position and dimension.

Given the association data of multiple point clouds, it becomes necessary to compare all of them at once to get a better overview of the final situation.

Figure 6
Visualisation of our difference detection operating on a point cloud and BIM model. Clutter and outliers are coloured pink, identical elements are grey, and elements which exist only in one model are colored red. Note that kinetic building elements (door leafs) are not yet detected in this version of the association algorithm. Source: Rislökka.

We tackle the problem of analysing multiple files at once, again using set theory. The simplicity of this approach enables the user to create his/her own detection rules for the filtering of building elements on the fly.

Step 4: Refinement of detection and feedback to user

The output of the analysis in the previous step indicates whether an element is present or not in the other representation (Figure 6).

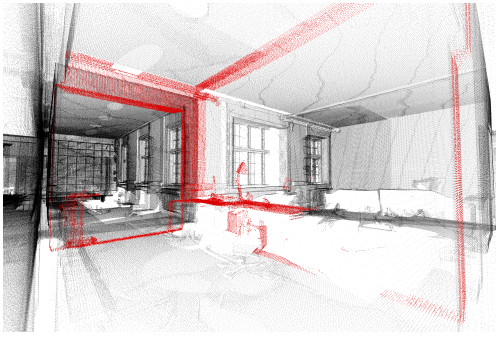


Figure 7
Point Cloud of a room which was extended by taking down two walls. The red points indicate the areas, where these were located and show as well, that the removal was only partial. An overview of the spatial situation is shown in Figure 8.

We find however that sometimes elements are only partially scanned or that parts of a building element are removed. Figure 7. shows for instance the case of walls, which have been removed except a part in the top, which remains as ceiling joist. In this case points can still be associated to the building element in the BIM representation and the algorithm would

indicate, that the element didn't change. We use additional data, which is delivered with the RDF file in order to refine the difference detection. The RDF association file contains information on the amount of points which theoretically should be associated with each of the BIM elements - the number of potential points scanned in perfect conditions. Comparing this measure with the actual number of points referred to at building element, allows us to deduct parts of the building which are missing from the point cloud. This also reveals the improperly scanned (obstructed) elements of the building. In these cases it can be indicated to the user, that a further visual inspection is required.

It is important to note that some changes can be improperly assigned to the point cloud, as caused by the noise in the data. Noise refers here to the instrument-inducted noise (whose value is given by the manufacturer) as well as points reflected from furniture or other non-related objects. It is necessary to perform a qualitative analysis of the association to identify its errors. By randomly sampling the point cloud and measuring the closest point distance to the associated geometry, we obtain the values of standard and average deviation. The data is compiled as a set of histograms (one for each associated pair), which serves as an assistance for an eventual inspection.

EVALUATION OF THE DIFFERENCE DETECTION

The functioning of the approach was positively evaluated on 3 real world data sets, of which one is described in the following paragraph. This set of data has one characteristic change to detect - the removal of two walls. A special difficulty is here, that only a part of the wall is removed (Figure 8), while a ceiling joist remains. It contains the ground truth (IFC file), a 1st scan (Time 1), which shows the same state of the building as the ground truth and two scans (Time 2 and 3) acquired over an extended period of time after the removal of the walls and finally an IFC file which includes additions and removals of elements. The algorithm detected the change from Time 1 to 2 truthfully and indicated no changes between Time 2 and 3.

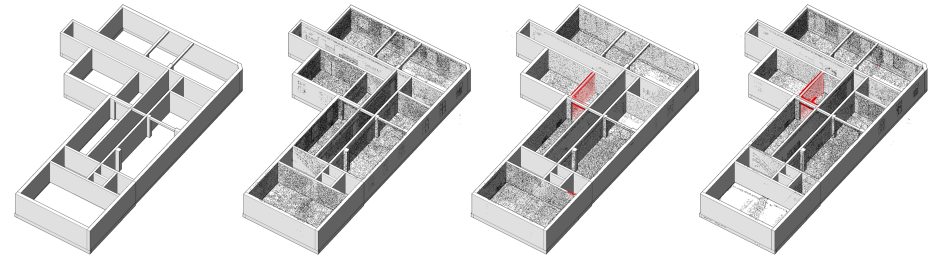
The previous evaluation demonstrates the validity of the approach. We extended the evaluation and tested, whether the approach is able to detect multiple changes (additions and removal of elements) over multiple files. For this we prepared a set of synthetic IFC files, which have the described characteristics in terms of changes. We execute our difference detection (Figure 9) and all changes are correctly detected.

DISCUSSION AND CONCLUSION

The rapid advance in 3D scanning and point cloud generating technologies as well as the proliferation of BIM in all areas of the building profession creates the foundation for a new information driven practice. It is based on data from sensors which provide a representation of the real world in almost real-time, and from linked databases, which hold the many sources of information that describe a building's performance, design intent, and specifications on all levels.

But with the vast amount of data that is accessible for institutions, architects, and engineers, comes an issue of data management and analysis. Various software tools developed in the DURAARK project try to simplify and automate currently tedious tasks. In this paper we presented the IFC reconstruction and Difference Detection algorithms and their implementation, in our case Grasshopper. The evaluation has proven the principal usability of the approach. An evaluation with further real world datasets should take place. A difference detection between different states of a building represented in Point Clouds on the base of reconstructed BIM files might sound tautologic in the first place. The use of the tool in practice examples demonstrated the advantages of the approach especially under four aspects:

1. A point cloud can be easily compared with existing BIM files.
2. Clutter in 3d scans, such as furniture and



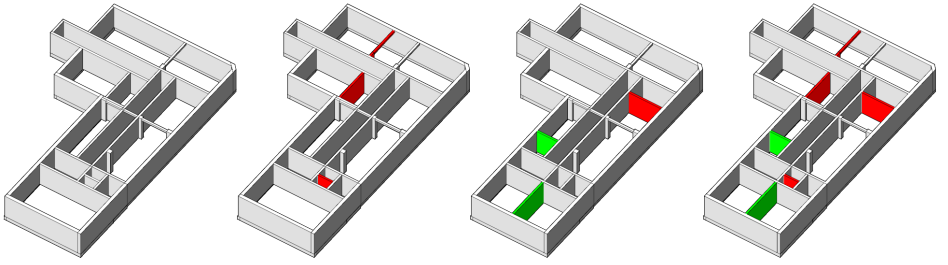


Figure 8
A ground truth IFC model (left) is compared with 3 point clouds acquired over an extended period of time. The removal of two walls (highlighted) is properly detected in the second and third scan.

Figure 9
Automatic analysis of changes between 3 IFC model of the same building. The ground truth (Time 0) is shown on the left. Removed elements are drawn in red, while additional elements are drawn in green.

scanned persons, is effectively filtered. Vice versa it provides a process of elimination of all points that are with a high probability identical between datasets and hence a smaller set to search for false negative etc.

3. Most importantly a hierarchy for the search is provided, which follows the common architectural understanding of building elements. The approach provides a mean to get an overview of differences on the level of the whole building, while a detailed investigation on the level of the element (for instance by means of a histogram of the distance between points points in certain distance to the building element) can provide numerical values to assess overall accuracy.
4. Indications for further search are provided: Being able to overlook a whole model allows users to identify critical areas, that might need further inspections.

Some issues need to be addressed in the future:

- The Difference Detection tool utilizes the data from the Association and Registration process to highlight the differences between data sets. The quality of the difference detection is therefore directly linked to the quality of the input data. The errors, which were exposed in the Difference Detection algorithm can be directly traced back to the input from the previous algorithms.
- The association of points to building elements

is the result of the preceding IFC reconstruction. The underlying algorithm detects currently only a narrow set of building elements. This should be extended to more building elements, like columns, heaters or any other infrastructural part in a building. The recognition or filtering of these would ease difference detection as noise is reduced. Additionally, the algorithm is currently limited to processing each story individually without taking their linkage via staircases or elevator shafts into account. The ability to differentiate really complex and data heavy point clouds of complete buildings remains a future task. A further possible extension for IFC reconstruction would be to recognize non-planar elements, like curved walls or arches.

- The faithful import of IFC files into the BIM modelling environment (Rhino with Visual Arq plugin) proved to be a challenge. This might be caused by the plugin, but points definitely at ambiguities inherent to the IFC format.

The approach presented in this paper allows to manage and process big data efficiently, further research is however required on fast and efficient noise removal, object recognition and the general precision in the point cloud processing.

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